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EDUCATION

Duration	Degree/Certificate	College/School	CGPA (10)/ Percentage
April 2017 - December 2022	PhD - Civil Engineering (Water Resources Engineering)	NIT Srinagar	8.0
August 2011 - August 2015	B.Tech (Civil Engineering)	IUST Awantipora	8.48
May 2009 - May 2010	10+2	CBSE	63.2
May 2007 - May 2008	10	CBSE	80.0

PUBLICATIONS

Journals

Fluvial Sediment Transport (Ph.D.)

- [1] **A. M. Bhat** and M. A. Ahanger, "Study of movability number as the parameter of threshold of sediment motion," *Modeling Earth Systems and Environment*, vol. 8, pp. 639–645, 2022. DOI: 10.1007/s40808-021-01122-7.
- [2] **A. M. Bhat**, M. A. Ahanger, and P. K. Mohapatra, "An examination of dimensionless variables in sediment threshold studies," *Geo-Marine Letters*, vol. 42, no. 1, pp. 6, 2022. DOI: 10.1007/s00367-022-00730-1.
- [3] **A. M. Bhat**, M. A. Ahanger, and P. K. Mohapatra, "Stability criteria for coarse sediment in natural streams," *Journal of the Geological Society of India*, vol. 99, no. 9, pp. 1285–1291, 2023. DOI: 10.1007/s12594-023-2462-2.
- [4] **A. M. Bhat**, M. A. Ahanger, and P. K. Mohapatra, "Movability number as the parameter of sediment incipient motion: A mathematical approach," *Ocean Science Journal*, vol. 58, no. 4, pp. 29, 2023. DOI: 10.1007/s12601-023-00112-3.
- [5] **A. M. Bhat** and M. A. Ahanger, "Influence of sediment grain protrusion on the movability number," *International Journal of Hydrology Science and Technology*, vol. 14, no. 2, pp. 124–139, 2022. DOI: 10.1504/IJHST.2022.124548.
- [6] **A. M. Bhat** and M. A. Ahanger, "Effect of non-uniformity of sediment size on the movability number," *International Journal of Hydrology Science and Technology*, vol. 14, no. 4, pp. 375–389, 2022. DOI: 10.1504/IJHST.2022.126428.
- [7] **A. M. Bhat**, M. A. Ahanger, and P. K. Mohapatra, "Threshold of motion of unconventional sediment under unidirectional flow," *International Journal of Hydrology Science and Technology*, vol. 15, no. 1, pp. 11–21, 2023. DOI: 10.1504/IJHST.2023.127890.

Urban Water Management (Postdoctoral Fellow)

- [8] **A. M. Bhat**, P. K. Mohapatra, and I. M. Tripathi, "Hydrologic-hydraulic response of swales to stormwater runoff," *Total Environment Advances*, vol. 12, 2024. DOI: 10.1016/j.teadva.2024.200118.
- [9] **A. M. Bhat**, P. K. Mohapatra, and I. M. Tripathi, "Towards green stormwater infrastructure: A review of the hydraulics of curb inlets in stormwater management," *Next Research*, 2025. DOI: 10.1016/j.nexres.2025.100251.
- [10] A. Sabut, **A. M. Bhat**, I. M. Tripathi, and P. K. Mohapatra, "Optimizing green infrastructure for urban flood mitigation using SWMM model: A case study in Kozhikode, India," *Journal of Hydrologic Engineering*, vol. 31(2), 2026, (Corresponding author). DOI: 10.1061/JHYEFF.HEENG-6677.
- [11] A. Sabut, **A. M. Bhat**, and P. K. Mohapatra, "Stormwater control measures, design aspects, and impact of climate change – a review," *Water-Energy Nexus*, vol. 9, 2025, (Corresponding Author). DOI: 10.1016/j.wen.2025.12.001.
- [12] I. M. Tripathi, S. Kumar, A. Modi, **A. M. Bhat**, C. Bhagat, P. K. Mohapatra, and M. L. Kansal, "Towards water sensitivity: A critical review of urban water management strategies," *Water Resources Management*, vol. 40, no. 3, pp. 1–25, 2026. DOI: 10.1007/s11269-025-04377-2.

- [13] **A. M. Bhat**, I. M. Tripathi, P. K. Mohapatra, and C. Bhagat, "Reimagining stormwater management: Sustainable drainage pathways for resilient Indian cities," *Environmental Conservation*, pp. 1–7, 2025. DOI: 10.1017/S0376892925100222.
- [14] **A. M. Bhat**, I. M. Tripathi, and P. K. Mohapatra, "How far have roadside curb inlets evolved towards sustainable urban drainage?" *Watershed Ecology and the Environment*, 2025. DOI: 10.1016/j.wsee.2025.05.003.
- [15] I. M. Tripathi, V. Jain, P. K. Mohapatra, **A. M. Bhat**, B. Parmar, A. Modi, and S. S. Mahto, "Impacts and consequences of reservoir on downstream river systems: Insights from the Ravi river in India," *Next Sustainability*, 2025, Under Review.
- [16] I. M. Tripathi, **A. M. Bhat**, A. Yadav, P. K. Mohapatra, and M. L. Kansal, "Swot analysis for sustainable stormwater drainage in indian cities," *Sustainable Geosciences: People, Planet and Prosperity*, 2026, Under Review.
- [17] **A. M. Bhat**, I. M. Tripathi, P. K. Mohapatra, and M. A. Ahanger, "Analytically supported empirical framework for runoff attenuation in vegetated swales," 2026, Under Preparation.

DOCTORAL THESIS

Analysis of Dimensionless Variables to Study the Problem of Sediment Threshold (Dr. A. M. Bhat, Dr. M. A. Ahanger and Dr. P. K. Mohapatra)

- Evaluation of bed configurational parameters on the sediment incipient motion in terms of Movability Number
- Development of an empirical model of sediment incipient motion in terms of Movability Number
- Study of sediment bed stability with various sediment threshold parameters
- Study of an alternative dimensionless sediment threshold parameter over the conventional Shields dimensionless shear stress and Movability Number

EXPERIENCE

- **Assistant Professor, NIT Srinagar, India** *April 2025 – till date*
 - Instructed core and elective courses for B.Tech and M.Tech students. Subjects taught include **Fluid Mechanics, Ground Water Hydrology, Environmental Engineering-I, Engineering Mechanics, Cost and Estimation in Civil Engineering**, and **Engineering Drawing**.
 - Supervised and conducted **laboratory courses** for B.Tech and M.Tech students, including the **Fluid Mechanics and Hydraulics Lab, Water Quality Lab**, and **Engineering Geology Lab**.
 - Mentored final-year B.Tech students, providing guidance on their major project work.
- **Postdoctoral Fellow, IIT Gandhinagar, Gujarat, India** *March 2023 - March 2025*
 - Studied stormwater management with roadside curb inlets and green infrastructure for sustainability.
 - Analyzed hydraulic-hydrologic performance of swales in stormwater mitigation.
 - Contributed opinion papers on stormwater management in India and abroad.
- **Teaching Assistant, NIT Srinagar, India** *July 2018 - July 2022*
 - Assisted in laboratory experiments for B. Tech and M. Tech students.
 - Conducted tutorial classes for B. Tech and M. Tech students.

TECHNICAL SKILLS

- **Laboratory & Experimental Experience:** Conducted a wide range of hydraulic and fluid mechanics experiments, including hydraulic jump, velocity distribution, discharge measurement, major and minor pipe losses, venturimeter, orifice meter, and flow over notches and weirs. Hands-on experience with laboratory equipment such as flumes, Double Ring Infiltrometer, Reynolds Apparatus, free and forced vortex setups, and Vectrino+ (ADV).
- **Software Proficiency:** MS Excel, GIS, HEC-RAS (2D), ANSYS Fluent, SWMM
- **Programming Skills:** MATLAB

REFERENCES

Dr. Manzoor Ahmad Ahanger

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